

An Experimental Testbed for the Performance Evaluation of Optical Time Difference of Arrival Based Indoor Positioning

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Abstract—Time Difference of Arrival (TDoA) based Visible Light Positioning (VLP) and Optical Wireless Positioning (OWP) offer considerable potential over conventional Received Signal Strength (RSS) methods. The current state of the art in TDoA OWP is however limited concerning experimental assessments, mainly focusing on simulations. This work presents experimental TDoA measurements in a proof-of-concept setup. The measurement results demonstrate a 95th percentile accuracy of 5.3 cm in a 50×50 cm area, in the absence of reflections. The susceptibility to reflections is examined experimentally, demonstrating that the performance deterioration can easily be minimized by performing a recalibration.

Index Terms—Experimental assessment, optical wireless positioning, reflections, time difference of arrival, visible light positioning.

I. INTRODUCTION

INDOOR positioning has been a longstanding research topic pursuing accurate and precise localization within indoor environments. Several different technologies have been developed to this end, including Visible Light Positioning (VLP), and more broadly Optical Wireless Positioning (OWP). These VLP and OWP systems employ optical signals within the visible light or Near-Infrared (NIR) wavelength region in order to enable indoor positioning. The aforementioned systems can be roughly classified depending on the measured signal property

and used positioning algorithm into Received Signal Strength (RSS), Angle-of-arrival (AoA) and Time-of-arrival (ToA) or Time-difference-of-arrival (TDoA) [1]. The vast majority of OWP systems make use of the RSS of optical signals in order to perform positioning [2], [3]. Although RSS-based solutions have been thoroughly studied, a number of inherent drawbacks are associated with this method. A tilt in the orientation of the transmitters and/or receivers can drastically decrease the performance [4]. Machine learning based methods are less susceptible to this performance deterioration by essentially embedding the impact of the transmitter tilt on the RSS in the model through the training dataset [5]. However, the acquisition of such a training dataset poses a cumbersome task in order to deploy this technology on a large scale. Additionally, ranging errors and subsequent positioning errors occur when the radiation patterns of the transmitters deviate from the ideal radiation pattern as defined by the specifications which are assumed within a channel model [6]. Machine learning methods can also mitigate this effect [7], essentially embedding the real radiation patterns in the model during the training phase. However, long-term changing effects should still be accounted for, such as LED aging [8], [9], dust suspended in the air [10] or accumulated on the transmitters and/or receivers [11], especially in industrial environments such as large warehouses [12]. Moreover, the inevitable need to replace faulty transmitters can be problematic considering that new LEDs cannot be guaranteed to have the same properties as the original transmitter on which the model was trained. All these long-term effects have an impact on the RSS values that is difficult to assess and compensate for accurately. Moreover, Non-Line-of-Sight (NLoS) contributions can, due to reflections, increase the RSS and thus negatively impact the performance of these methods. Although this impact can also be embedded in machine learning models through the training data in the case of static environments, dynamic changes in the environment are still difficult to compensate for, and frequent retraining of the models to account for changes in the environment is undesirable, if feasible at all. An extreme case requiring even more data acquisition is known as fingerprinting. This method measures the RSS or other relevant signal properties for a large number of locations and stores these fingerprints in a database [13], [14]. The location estimations are made by matching new measurements to the stored fingerprints.

Manuscript received 15 February 2024; revised 16 May 2024; accepted 5 July 2024. Date of publication 10 July 2024; date of current version 2 December 2024. This work was supported in part by the Central Europe Leuven Strategic Alliance (CELSA) Project CELSA-22-205: Positioning by means of Optical Wireless time Differences (POWER) and the Czech Technical University (CTU) Global Postdoc Fellowship Program, and in part by COST Action CA19111 (European Network on Future Generation Optical Wireless Communication Technologies, NEWFOCUS), COST (European Cooperation in Science and Technology). (Corresponding author: Jorik De Bruycker.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JLT.2024.3426033>.

Digital Object Identifier 10.1109/JLT.2024.3426033

The aforementioned disadvantages of RSS-based solutions form a major driver in researching AoA and ToA or TDoA based methods, as these positioning approaches are expected to be more resilient to these flaws, albeit in exchange for more complicated hardware requirements. AoA utilizes angular diversity receivers in order to retrieve the incident angle of several optical transmitters [15]. These receivers resolve a number of issues related to RSS methods, as addressed above. Yet the performance is still susceptible to the tilting of the receivers which can be challenging to compensate for.

ToA and TDoA measure the time and time difference respectively that it takes for an optical signal to travel between one or more transmitters and receivers. From several surveys on VLP and OWP [2], [3], it is noteworthy that ToA and TDoA based methods are uniquely included in studies focusing on simulations. Recently, however, a small number of research groups have demonstrated experimental measurement setups and results employing ToA/TDoA based approaches in OWP scenarios. Researchers at Fraunhofer HHI have employed Orthogonal Frequency-division Multiplexing (OFDM) hardware based on the ITU-T G.9991 standard [16] to perform experimental ranging measurements [17]. The frame synchronization and channel estimation preamble were used to perform a coarse and fine timing measurement, respectively. The results demonstrated ranging errors between transmitters and receivers below 5 cm. The subsequent work of the same authors expanded hereupon by performing positioning in a $1 \times 1 \times 2$ m volumetric setup using 4 fixed transmitters [18]. The positioning was then evaluated for a total of 12 different points, for which the results showed a mean Euclidean positioning error below 8 cm. Their following work enlarged the measurement setup to include 6 transmitters covering a 1.5×3.5 m rectangular area [19]. The authors proposed an optimization approach to correct for a systematic angle dependency of the used hardware. The proposed method improved the performance by decreasing the positioning Root-mean-square Error (RMSE) from 7, 6 and 3 cm to 5, 4 and 1 cm on the x -, y - and z -axis, respectively. The research group at Center for Sensor Systems (ZESS) presented work using existing Time-of-flight (ToF) cameras, which are repurposed as a receiver for positioning [20]. By means of the spatial resolution of the ToF cameras, both AoA and ToF information was acquired. An AoA/TDoA fusion algorithm was proposed and the performance was compared to existing methods. Experimental acquisition of amplitude and depth images was demonstrated for a single transmitter in a $2 \times 2 \times 2.5$ m measurement setup and further simulation under typical noise conditions demonstrated that the proposed method could improve the RMSE Euclidean positioning error to 2.8 cm. Related work from ZESS presented a novel solution for pseudo-passive ToF imaging [21]. By employing a separate VLC transmitter module from the ToF camera module, the power hungriness issue of ToF cameras is resolved. The authors employed a reference photodiode to retrieve an external reference signal for the photonic mixer device (PMD) camera. The PMD hardware implements cross-correlation of this reference signal on a pixel level scale, enabling accurate depth reconstruction of a camera scene. Preliminary 3D imaging results demonstrated a worst case depth error of 20 mm on a

25 cm distance of the camera target. Another research group presented a novel approach relying on differential phase of arrival [22]. The authors deployed 4 transmitters serving as a dual-frequency orthogonal beacon. They derived differential phase information at the receiver in order to perform the position estimation. Their simulation results demonstrated a performance increase compared to Time-difference-of-flight (TDoF)-based schemes with a 90th percentile accuracy of 1.3 cm, meaning 90 % of the results have an error smaller than 1.3 cm. Additionally, an experimental measurement setup was constructed, demonstrating an average positioning error of 7.12 cm on a 3×3 m grid.

In this work, we explore the feasibility of performing TDoA positioning by measuring time differences between received laser pulses. These pulses are emitted by a NIR laser with a radiation pattern with a wide Full Width at Half Maximum (FWHM) and measured at multiple fixed receivers with known positions. On the transmitter end, high current pulsed lasers are used, emitting pulses that can be modeled as a Gaussian pulse in the time domain. Similar pulse shapes are employed in the field of Light Detection and Ranging (LiDAR), where the design of a time pick-off circuit has been simulated and experimentally assessed [23] and the impact on ranging accuracy and precision caused by this pulse shape has been studied in detail [24]. The aforementioned studies make use of directional lasers to measure one-dimensional distances to a reflective surface along the laser axis up to distances of 100 m.

In this work, we examine the possibility of using both transmitters and receivers with a wide FWHM radiation pattern and a wide Field of View (FoV), respectively. The content is structured as follows. Section II describes the hardware used and discusses the associated pulse shape and signal processing. A first evaluation of the ranging performance of the selected hardware and signal processing is performed in Section III. In that Section, the time differences are measured for two receivers and the results are compared for two different tier oscilloscopes. Thereafter, Section IV presents TDoA positioning measurements and results using four fixed receivers.

An experimental assessment of the impact of reflections is described in Section V. The results are further elaborated and discussed in Section VI. Finally, Section VII summarizes the main results and contributions of this work.

II. HARDWARE AND GAUSSIAN PULSE FITTING

The experimental setup for TDoA positioning, developed in this work, utilizes a single mobile transmitter, and multiple synchronized receivers with a fixed, known position, as illustrated in Fig. 1. This transmitter consists of an OSRAM V205Q131A-940 Vertical Cavity Surface Emitting Laser (VCSEL) operating at 940 nm mounted on an EPC9126 Resonant Capacitive Discharge Laser Diode driver. The radiation pattern of the VCSELs illuminates at a FWHM of 78° by 65° . The aforementioned driver emits pulses shaped by the sinusoidal ringing from a resonant network formed by a discharging capacitor bank and the total power loop inductance dominated by the parasitic inductance of the mounted laser diode. In theory, the resulting current pulse should correspond to a single positive half of a sine

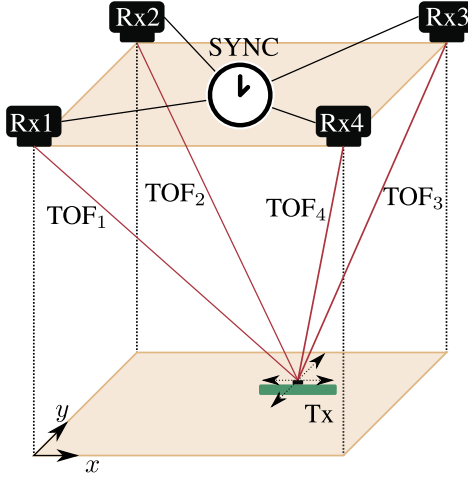


Fig. 1. Overview of the experimental TDOA measurement setup.

wave. In the LiDAR literature, the received pulse shape is also often modeled as a Gaussian pulse shape, or as a heavy-tailed or a truncated parabola [23].

The receiver side uses a Thorlabs PDA10A2 fixed gain Silicon photodetector. This detector features a photosensitive surface of 0.8 mm^2 with an analog bandwidth of 150 MHz and a transimpedance gain of 5 kV/A at a 50Ω termination and has a responsivity of approximately 0.125 A/W at the laser wavelength of 940 nm. Photodiodes exhibit a fundamental trade-off between gain and bandwidth [25], [26]. Hence, the selected photodiode is considered suitable, as a higher bandwidth alternative would sacrifice gain and thus achievable range. The received signals are the result of a convolution of the impulse responses of the transmitter and receiver hardware and the channel impulse response according to:

$$r(t) = h_{tx}(t) * h_c(t) * h_{rx}(t) * s(t), \quad (1)$$

where $h_{tx}(t)$, $h_c(t)$ and $h_{rx}(t)$ denote the impulse response of the transmitter, the channel and the receiver respectively. $s(t)$ and $r(t)$ denote the transmitted and received signals respectively.

In this work, the received signal $r(t)$ is modeled as a Gaussian pulse $r_G(t)$ of order n given by:

$$r_G(t) = A \cdot e^{-\left(\frac{(t-t_0)^2}{\sigma^2}\right)^n} + B, \quad (2)$$

where A denotes the pulse amplitude, t_0 the time offset corresponding to the pulse peak time, σ and n affect the pulse width and order respectively, whereas B denotes a DC-offset of the pulse. This model extends upon [24] by including the n to account for a higher order Gaussian. For the purpose of measuring time differences, t_0 is naturally the most important parameter in this model. The absolute value for t_0 is arbitrary, yet the differences in the values of t_0 can be determined for the signals received by multiple synchronized receivers. When multiplied by the propagation speed, these time differences correspond to the path length differences between the transmitter and the receivers in question. It is important to notice that the received pulses are the result of the impulse response function of the full system, mainly dominated by $h_{rx}(t)$ due to the lower analog

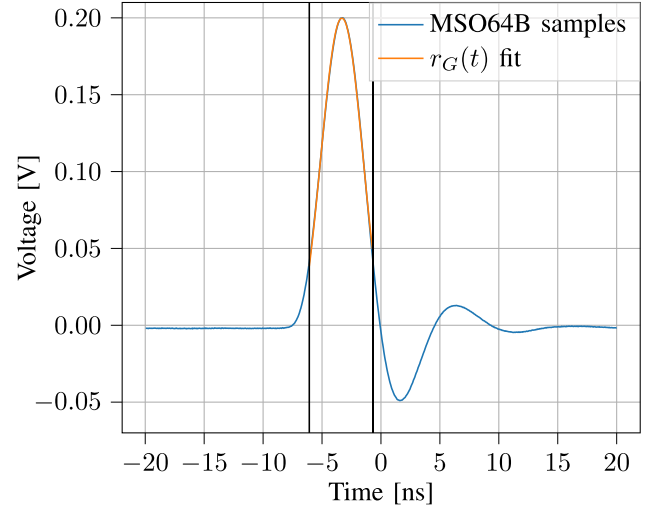


Fig. 2. Example pulse shape measured with MSO64B sampled at 25 GS/s sample rate or 40 ps time resolution.

bandwidth of the photodetector of 150 MHz. This bandwidth limit results in the time domain spreading of the pulse. Despite this effect, the time differences between two photodetectors can still be estimated with sufficient accuracy for the purpose of this work. In order to verify and ensure the validity of the Gaussian pulse shape model, an initial assessment of the received pulse shape is made using a high-end Tektronix 6 series MSO64B oscilloscope. This scope provides 10 GHz analog bandwidth and can be used to sample 4 channels simultaneously at a sample frequency f_{sample} of 25 GS/s, corresponding to a time resolution T_{sample} of 40 ps. To reduce the impact of the noise, the scope is set to average a total of 1024 acquired pulses. An example of an acquired pulse is shown in Fig. 2. The selected hardware results in a received pulse width typically in the order of 3 to 4 ns FWHM. The acquired signal also shows a typical overshoot, this can be attributed to the higher order filter response of the full receiver chain. t_0 can be retrieved by performing a Least Squares Error (LSE) function fit on the positive end of the acquired pulse. The interval of samples on which the fit is performed is indicated by the vertical lines in Fig. 2. The constraints determining this interval are determined as follows. The discrete samples of the acquired signal are denoted by $r[k]$, where k is the sample index and $r[k]$ represents the value of $r(t)$ at a discrete time step $k \cdot T_{\text{sample}}$ with $T_{\text{sample}} = \frac{1}{f_{\text{sample}}}$. The sample index for which $r[k]$ reaches its maximum value is denoted as k_{max} :

$$k_{\text{max}} = \arg \max \{r[k]\}. \quad (3)$$

The bounds for the Gaussian pulse LSE fit calculation can then be noted as $[k_{\text{start}}, k_{\text{stop}}]$, defined by:

$$k_{\text{start}} = \min_k [r[k]] - 1 \quad \text{s. t.} \quad r[k] > p \cdot r[k_{\text{max}}] \quad (4)$$

and

$$k_{\text{stop}} = \min_k [r[k]] \quad \text{s. t.} \quad r[k] < p \cdot r[k_{\text{max}}] \text{ and } k > k_{\text{max}} \quad (5)$$

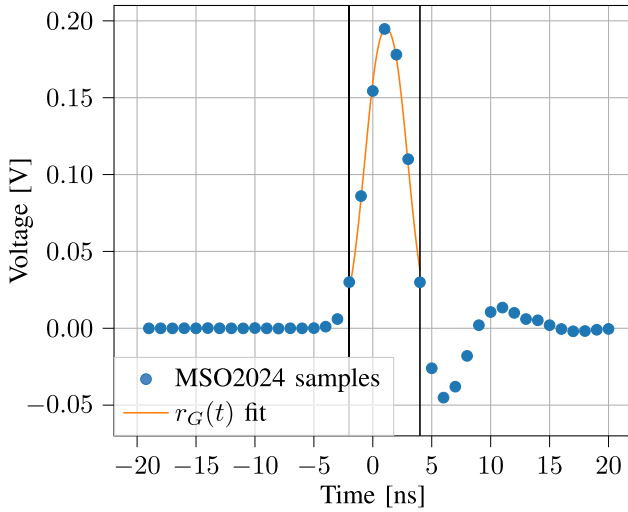


Fig. 3. Example pulse shape measured with MSO2024 at 1 GS/s sample rate or 1 ns time resolution.

Here k_{start} , corresponds to the last sample before the signal reaches a certain threshold value of $p \in [0, 1]$ times the maximum value of the signal. k_{stop} resembles the first value that falls below the threshold value of p times the maximum value of the signal after the signal peak. Initially, p is set to a value of 0.2 (= 20%). For lower pulse amplitudes however, determining these bounds for the pulse fitting based on this method can lead to incorrect bounds or even failure to determine the bounds due to noise distorting the pulse shape. In this case, the threshold percentage p is iteratively incremented by 0.01 (= 1%) until a correct fitting can be performed. To analyze the performance for a middle-end oscilloscope, identical tests were performed with a Tektronix MSO2024 oscilloscope for comparison. This scope is an order of magnitude lower in price compared to the MSO64B. The MSO2024 scope is set to acquire the maximum available periods to average of 512. The acquired samples from the MSO2024 oscilloscope are depicted in Fig. 3, these are plotted as single markers to clearly display the limited samples used due to the lower sampling rate. The MSO2024 oscilloscope provides 200 MHz analog bandwidth and is able to sample at $f_{\text{sample}} = 1 \text{ GS/s}$, corresponding to $T_{\text{sample}} = 1 \text{ ns}$ time resolution. The main performance difference between the MSO2024 and the MSO64B oscilloscope for the purpose of this work is the lower maximum sampling rate available. The analog bandwidth provided by the MSO2024 is significantly lower. Nevertheless, this is considered sufficient as the used receiver is already restricting the bandwidth to 150 MHz. The lower sampling rate and corresponding coarser time resolution of 1 ns do however complicate the accurate determination of the path length differences for multiple receivers. The Gaussian pulse LSE function fit previously discussed can calculate a t_0 value independent of the 1 ns time resolution. This approach thus provides sub-resolution results, which are vital to provide positioning with an acceptable resolution. The received signals are namely too coarsely sampled in the time-domain in order to provide enough resolution using a typical cross-correlation approach. In that case, the cross-correlation of two signals

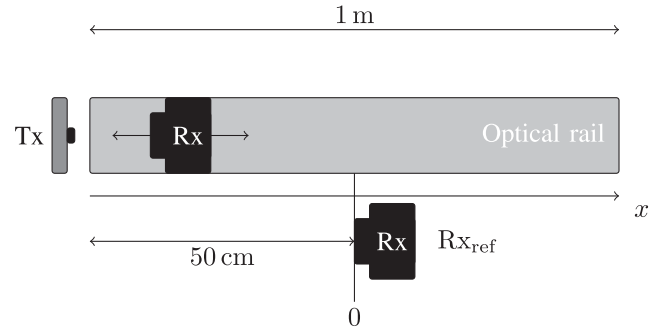


Fig. 4. Measurement setup used to compare measured distance differences between 2 receivers. Rx_{ref} is used as the reference and is positioned at a fixed lateral distance of 50 cm while the other moves along the x -axis in 5 cm increments from -30 cm to 45 cm with respect to Rx_{ref} .

is calculated in order to retrieve the time delay between the signals, and thus the corresponding path length difference. The maximum of the calculated cross-correlation then indicates the time delay. This method however limits the resolution of the calculated time delays to the time resolution of 1 ns. As this corresponds to a distance difference of 30 cm at the speed of light, this is considered insufficient for accurate TDoA positioning.

III. MODEL EVALUATION WITH INITIAL TIME DIFFERENCE MEASUREMENTS FOR RANGING

In order to first evaluate the performance of the Gaussian pulse fitting method for both oscilloscopes, a simple measurement setup is constructed which is illustrated in Fig. 4. Two identical receivers are connected with identical cables and connectors to the oscilloscope. One of these receivers is used as the reference, denoted as Rx_{ref} , and is mounted at a fixed lateral distance of 0.5 m from the transmitter. Note a small lateral offset from the optical rail of 5 cm in the actual setup. Further measurements take into account the actual hypotenuse distance between Tx and Rx. The other receiver is mounted on a 1 m optical rail and is moved along the x -axis in 5 cm increments from -30 cm to 45 cm , with Rx_{ref} corresponding to the zero point of the x -axis. Fig. 5 illustrates the received signals acquired by the MSO64B for Rx_{ref} and for the moving receiver at $x = -30 \text{ cm}$ and $x = 30 \text{ cm}$ as an example.

The Gaussian pulse fitting as described in Section II is used to retrieve the t_0 for all signals with p initialized to a value of 0.05. Even though all receiver hardware and cabling are supposedly identical, the fitting results however indicate a consistent erroneous offset for the difference of the t_0 values with respect to Rx_{ref} . This bias is attributed to a combination of oscilloscope channel skewing, which has not yet been accounted for, and an inevitable small asymmetry on the receiver hardware due to tolerance margins on the specifications. At this nanosecond range time scale, this effect is however non-negligible and should be compensated for, which is executed in post-processing in the form of a calibration. This calibration is performed easily by acquiring the t_0 value for both receivers at a known distance. The t_0 value is obtained for both the fixed receiver at $x = 0 \text{ cm}$ and the moving receiver at $x = -30 \text{ cm}$. This position is chosen as the intensity of the received pulse is the highest here and

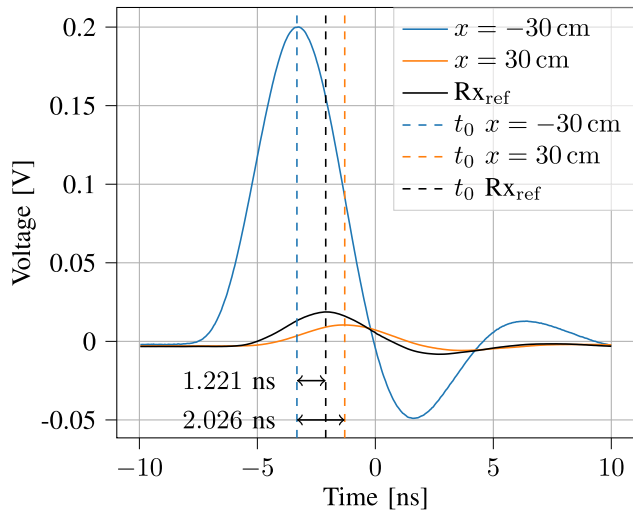


Fig. 5. Received signals at $x = -30$ cm and $x = 30$ cm compared to the Rx_{ref} signal. The t_0 values from the Gaussian pulse fittings are marked. The small time skew between both receivers will be compensated for in post-processing.

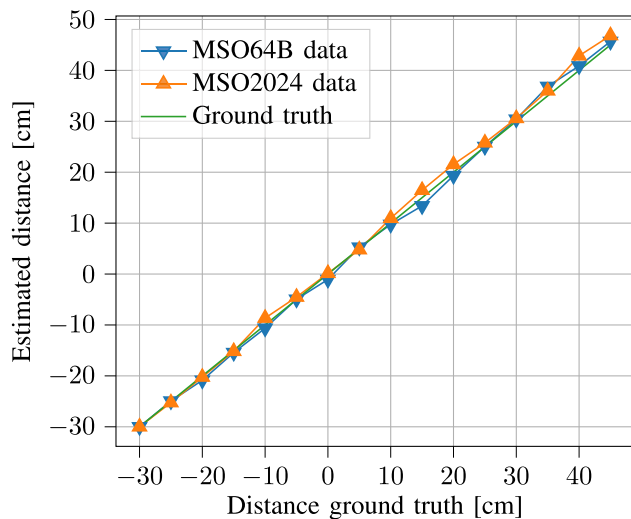


Fig. 6. Distance estimation from Gaussian pulse fitting from data acquired with MSO64B and MSO2024 oscilloscopes.

thus the pulse fitting is expected to be most reliable in this case. The difference between the measurement result and the ideal ground truth value is subtracted from subsequent measurement results as a calibration step. This way, not only the oscilloscope channel skew is calibrated, but also asymmetric delays in the full receiver chain. Note that a better calibration procedure could be performed if more measurement positions are considered. The described procedure, however, has the advantage that is very simple and practically applicable with only a single measurement position. With this calibration performed, the distance differences can be calculated and evaluated. The results are displayed in Fig. 6. The error on the measurement results is shown in Fig. 7. Two important conclusions can be drawn from studying these plots where the two oscilloscopes are being deployed consecutively. Firstly, the error is marginally higher at the highest distances with respect to the transmitter, explicable

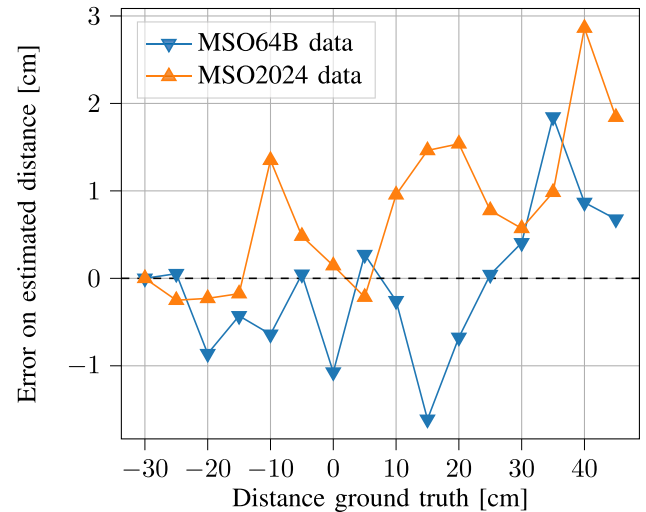


Fig. 7. Error on the distance estimation from Gaussian pulse fitting from data acquired with MSO64B and MSO2024 oscilloscopes.

by the lower signal levels. Secondly, the measurements from both oscilloscopes demonstrate comparable errors. Although the MSO2024 shows some slightly higher errors, this oscilloscope is considered as an adequate and cost-efficient solution for further experiments and will thus continue to be used in the following measurements.

IV. TIME DIFFERENCE OF ARRIVAL POSITIONING MEASUREMENTS ON A GRID

In order to obtain a large number of measurements for TDoA positioning evaluation on a grid in an accurate and practical manner, the transmitter is mounted on an OpenBuilds ACRO 1515 60"×60" motorized XY-stage. This allows for sub-mm accurate movements on a grid of 130×130 cm. The measurement setup is shown in Fig. 8, where the V205Q131A-940 VCSEL transmitter can be moved accurately across the full measurement plane indicated. The VCSEL is oriented with the surface normal pointing vertically upwards, thus along the z -axis in the right-handed coordinate system of Fig. 8. As the radiation pattern is asymmetric, it is important to notice that the VCSEL is oriented with the widest FWHM angle of 78° along the x -axis and the FWHM angle of 65° along the y -axis. Parallel to the measurement plane, four PDA10A2 photodetectors are fixed on the corners of a 1×1 m square surface and connected to the MSO2024 oscilloscope, using identical cables and connectors. The receiver plane is located at a height of 124 cm with respect to the VCSEL transmitter. With the aforementioned FWHM of 78° by 65° , this means the transmitter illuminates a rectangular surface of approximately 201×158 cm in the receiver plane, corresponding to a maximum distance in the receiver plane of 100 cm along one axis and 79 cm along the perpendicular axis in which the photodetectors are within the FWHM. A grid of 50×50 cm is measured in steps of 2.5 cm, resulting in a total of 441 measurements.

Fig. 9 illustrates the top view of the measured grid and the receiver positions in the XY robot coordinate system. As noted

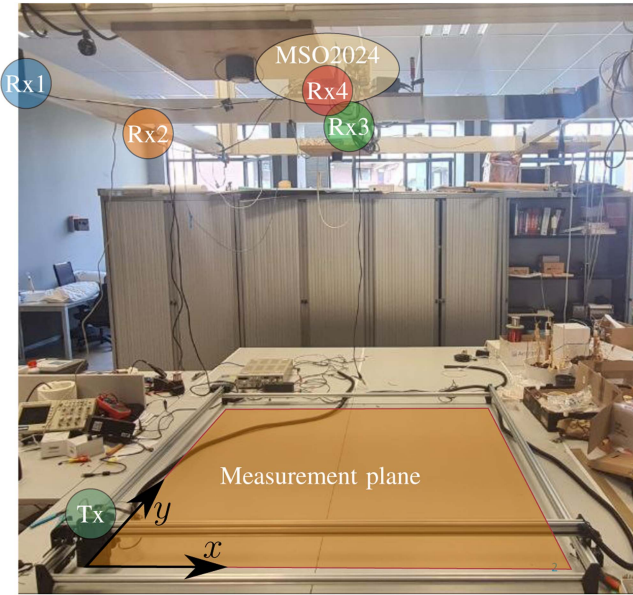


Fig. 8. Photograph of the automated measurement setup with all hardware indicated.

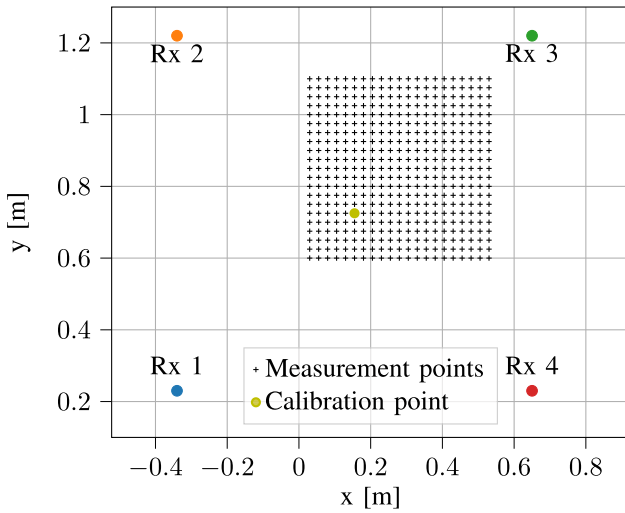


Fig. 9. Measurement points in the XY robot coordinate system. The calibration is performed using the center point of the grid as indicated with the yellow dot.

earlier, a calibration is required to compensate for time skew between different receivers and oscilloscope channels. This calibration is performed based on the measurement results with the transmitter located in the center of the grid, indicated by the yellow dot in Fig. 9. The received pulses for all receivers measured at this point are shown in Fig. 10. It is clear that these signals are not identical as would be expected in the ideal case. This is likely attributable to the combination of a small tilting of both the transmitter and receivers, minor asymmetry of the VCSEL radiation pattern, and a slight rotation of the transmitter around its normal axis due to manual soldering misalignment. Although this effect causes the received amplitudes to vary slightly, especially noticeable at Rx3, these effects have no impact on the ToF nor the t_0 calculated by the pulse fitting.

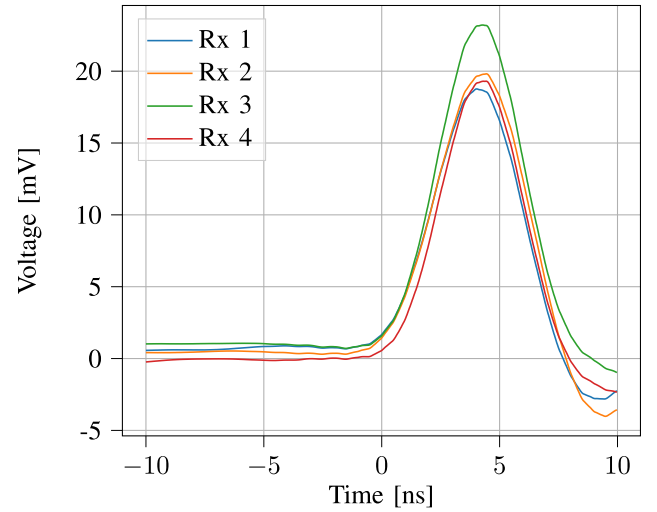


Fig. 10. Measured pulses at the four receivers when the laser is positioned in the center of the grid.

TABLE I
CALIBRATION MEASUREMENT RESULTS AT CENTER OF THE GRID

	Time difference t_0 [ps]	Path length difference [cm]
Rx2 - Rx1	148.1	4.44
Rx3 - Rx1	155.4	4.66
Rx4 - Rx1	188.4	5.65

All time and distance differences are referenced with respect to Rx1.

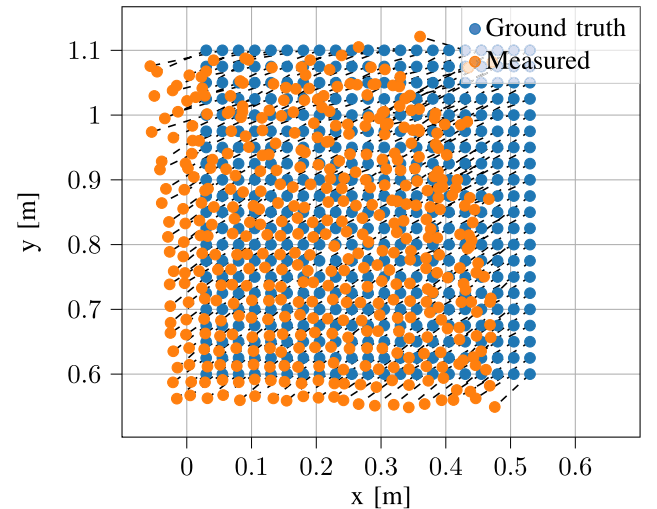


Fig. 11. TDoA positioning results using uncalibrated measurements.

This observation clearly contrasts the challenges of RSS-based positioning.

The time differences from the pulse fitting t_0 are calculated referenced to Rx1 and are shown in Table I. These values confirm the time skew and thus corresponding measurement offset for the associated path length difference. The calibration is further executed in post-processing by subtracting the offsets measured at the center of the grid from all subsequent measurements. This is also clearly visible from the results in Fig. 11, where the TDoA positioning is calculated without calibration from the

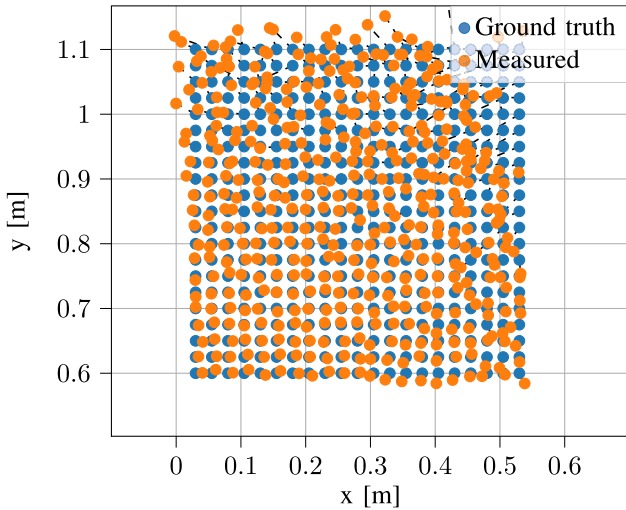


Fig. 12. TDoA positioning results after calibration.

measured path length differences associated with the difference in t_0 for all Gaussian pulse fitting results with p initialized to a value of 0.05. The TDoA positioning calculation is performed by an LSE optimization minimizing the error of the measured path length differences. While all receivers are coplanar in this configuration, the unique determination of 3D positioning is theoretically infeasible, even with four beacons [27]. Nevertheless, 2D positioning already serves a vast number of practical use cases [28]. This work thus focuses on 2D positioning in the xy -plane, achieved by constraining the z -component to the known height of 1.24 m to calculate path length differences. The ground truth locations are indicated with the blue dots forming the grid, while the associated calculated positions are represented with the orange dots, and connected to their respective ground truth position. The positioning results evidently show an offset in the direction of Rx1, in accordance with the overestimated path length differences for Rx2 through Rx4 demonstrated by the calibration measurement. After calibration, the positioning results improve significantly as the offset is removed, this is clearly seen in Fig. 12. Due to the limited FWHM of the VCSEL, the error on the positioning increases towards the upper right corner of the grid near Rx3. The pulse amplitude reduces drastically here for both Rx1 and Rx4, as these are positioned near the edge of the VCSEL radiation pattern. The cumulative distribution function of the positioning error in Fig. 13 presents the positioning performance of the TDoA both before and after calibration. The P95 (95th percentile) positioning error after calibration results in 5.3 cm with an RMSE of 2.5 cm. As a comparison, the results with some alternative methods for time difference estimation are also plotted in this figure. Simple peak detection (PD) quantizes the distance estimation to 30 cm resolution, thus leading to unsatisfactory results. Peak detection after sinc interpolation improves the performance due to the finer time and distance difference resolution, yet it still proves to be too unreliable for accurate positioning. For both these methods, the error is so significant that a similar calibration as previously discussed is unreasonable to improve the performance. Finally, cross-correlation after sinc interpolation results in slightly inferior results compared to the

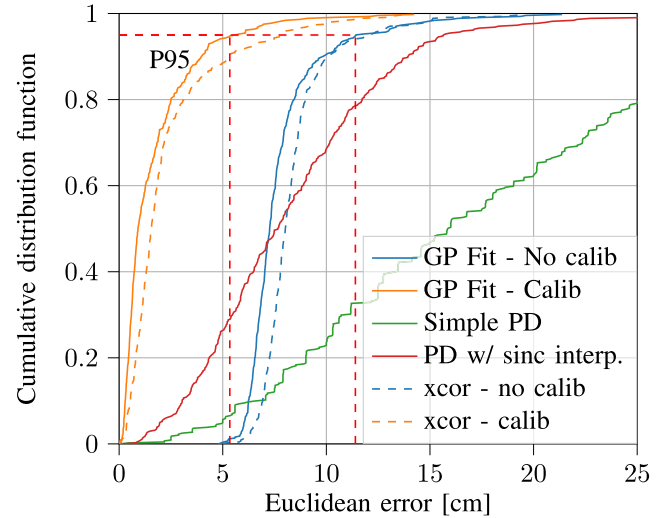


Fig. 13. Cumulative distribution function of the localization error for the uncalibrated and calibrated cases of GP fitting, compared to alternate time difference estimation methods.

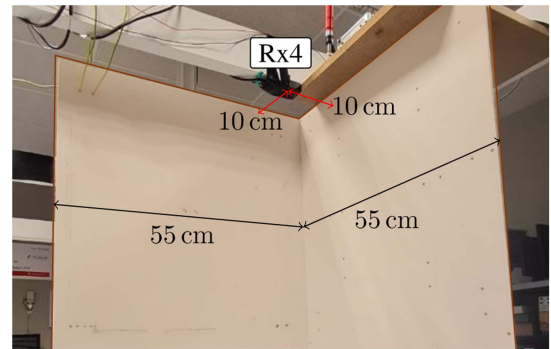


Fig. 14. Measurement setup with white panels mounted in the corner at Rx4.

Gaussian pulse fitting, also requiring an identical calibration procedure. The best performance is achieved by employing the Gaussian pulse fitting method. For this method, prior knowledge of the pulse shape is essential in order to select an accurate fitting model, enabling an accurate estimation of the time differences, even if a limited number of samples is available.

V. EXPERIMENTAL ASSESSMENT OF THE IMPACT OF REFLECTIONS

The aim of this Section is to assess the impact of NLoS contributions if the previously described positioning method is deployed. Related work dealt with this multipath interference by modifying the system model [29], [30]. This work, however, focuses on evaluating the robustness of the proposed simple Gaussian pulse model when multipath interference is initially left unaccounted for. In order to introduce reflections in the setup, white panels are fixed near a corner as shown in Fig. 14. The longer path length associated with these reflected NLoS contributions will cause a spread of the received pulses in the time-domain. The white panels are only installed near one of the receivers, namely Rx4. The panels are spaced in a corner at a

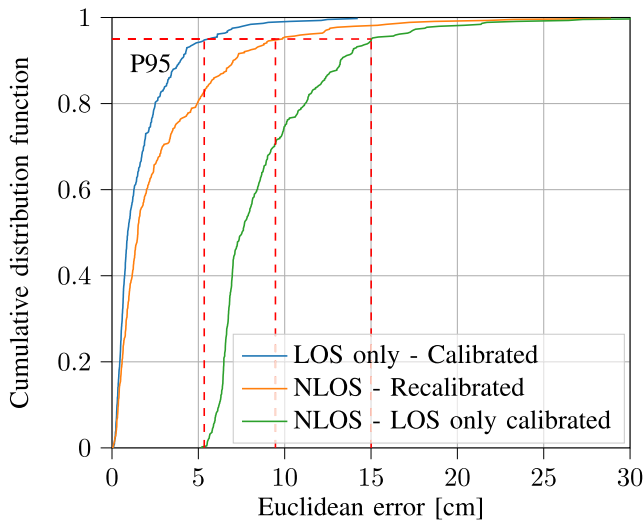


Fig. 15. Cumulative distribution function of the localization error for Line-of-Sight (LoS) case, and the case with reflections at Rx4. In the NLoS case, a distinction is made between unaltering the calibration from the LoS case, and performing a new calibration based on a single measurement in the center of the grid.

distance of 10 cm in both x and y directions. Fig. 14 displays a picture of Rx4 with the white panels in place, both 55 cm wide. This corner geometry can be considered as a worst case scenario where only a single receiver suffers an increase in its measured path length difference due to the time-domain spreading of the pulse.

Initially, the existing calibration from Table I without reflections is maintained, while the position estimations are calculated from new measurements including reflections. The cumulative distribution function of the positioning error for this case with reflections compared to the original LoS only case is presented in Fig. 15. The performance of the positioning method elaborated in previous Sections is impacted by two main differences that are apparent in this graph. Firstly, the curve flattened out, which implies a larger spread of the positioning errors. Secondly, the curve has an offset of around 5.2 cm. The latter is caused by the systematic overestimation of the time difference measured at Rx4, resulting in the calculated positions being skewed away from Rx4. Subsequently, a recalibration is performed by once again acquiring the time differences from a measurement at the center of the grid, but this time with the white panels mounted near Rx4. This recalibration leads to an improvement of the positioning performance as also indicated in Fig. 15. Compared to the case with reflections where the original calibration is maintained, performing a recalibration causes the P95 value to improve from 15.0 cm to 9.5 cm while the RMSE decreases from 9.5 cm to 4.6 cm. This recalibration is convenient to execute in practice requiring only a single new measurement. However, it is important to notice that this method does not account for the time-domain spreading varying spatially across the grid, as it assumes a new, constant time offset for all receivers. The results are thus mainly aimed to illustrate the impact of the performance of the described method when reflections are introduced, without altering the calibration procedure. In practice, the recalibration is

only required if significant changes are made to the environment, which is expected to be infrequent.

VI. DISCUSSION

The presented measurement setup and results demonstrate the viability of TDoA based positioning solutions on a lab scale.

Table II gives an overview of the performance metrics of this work in comparison to results reported in the related work mentioned in Section I on T(D)oA positioning. In summary, the following advances with regard to the current state of the art can be identified. Firstly, the results demonstrate both better RMSE and mean Euclidean error, although on a smaller scale, confirming the potential of the proposed system. Additionally, the number of evaluation points for a photodiode based system is the largest currently reported in experimental assessments, giving increased insights into the spatial distribution of the position estimation errors and the root cause associated with them. While the positioning results show a high accuracy with a P95 value of 5.3 cm, several challenges should be addressed to further increase the achievable coverage area for practical applications. Firstly, the coverage area can be improved by employing a different VCSEL transmitter with an even wider radiation pattern. Widening the pattern does however reduce the incident optical power on the receivers as a similar emitted power is distributed over a larger solid angle. In order to still receive the pulses with a high enough intensity to perform an accurate and reliable pulse fitting, the currently used photodetector modules have insufficient photosensitive area. This challenge can be addressed by designing a dedicated receiver with a larger photosensitive area. Increasing the surface area exhibits a trade-off, negatively impacting the receiver bandwidth due to increased junction capacitance [25]. Particular attention is thus required in the design of the transimpedance amplifier to maintain a sufficiently high analog bandwidth. Special techniques such as bootstrap transimpedance amplifier designs can enable this requirement [26]. The aforementioned improvements are expected to enable a similar setup with 4 photodiodes to cover a larger area. Nevertheless, accurately determining the optimal number of receivers needed to cover a specific area proves challenging and is considered beyond the scope of this study.

In order to cover even larger areas, additional receivers will be necessary. In essence, the synchronization of all receivers can still be performed by wiring all of them to a central acquisition system. It is important to correct for any possible cable length mismatches by calibrating the channel skew as illustrated in this work. For even larger areas, several positioning cells can be created, each covered by a limited number of receivers per acquisition system.

The operation of the proposed method benefited from averaging several acquisitions on the MSO2024 oscilloscope. Although the pulse repetition frequency of the Tx can be sufficiently high, the processing on the scope introduces additional delay between subsequent acquisitions. This limits the acquisition rate per second and results in approximately 30 s latency per measurement of the 512 averages acquired. The current implementation is thus unable to support a mobile scenario. This limitation is,

TABLE II
COMPARISON TABLE OF THE PERFORMANCE METRICS OF DIFFERENT WORKS PERFORMING T(D)OA-BASED POSITIONING

Reference	Dimensions	Rx type	Number of Beacons	Evaluation points	x	y	z	Euclidean		
					RMSE [cm]			Mean [cm]	P95 [cm]	
[18]	$1 \times 1 \times 2$ m	PD	4	12	5	4	1	≤ 8		
[19]	1.5×3.5 m	PD	6	35				3.3		
[20]	$2 \times 2 \times 2.5$ m	ToF Camera	5	400				2.8		
[22]	$3 \times 3 \times 3$ m	PD	4	36				7.12		
Our work - LOS	$0.5 \times 0.5 \times 1.24$ m	PD	4	441	2.1	1.4	-	2.5	0.9	5.3
Our work - NLOS					4.1	2.1	-	4.6	1.5	9.5

Empty values are not reported in the referred publications.

however, fully attributable to the practical implementation. Enabling mobile scenarios is feasible by utilizing dedicated analog hardware or a Field-programmable Gate Array (FPGA) in order to accelerate the acquisition and processing. Additionally, sensor fusion with inertial measurement unit data can further improve performance for mobile applications [31].

3D positioning is not considered in this work, as performing TDoA positioning with 4 co-planar receivers leads to ambiguous location estimations. For example, at the center of the grid, all distances between Tx and Rx are identical. The range differences are thus zero. This is, however, true for all Tx heights (z -component) at these (x, y) coordinates. Enabling 3D positioning thus necessitates resolving this ambiguity. One possible way in which this can be achieved is by utilizing amplitude information in conjunction with the TDoA measurements.

Despite the aforementioned challenges, the proposed positioning method has numerous advantages over typical RSS-based implementations. A number of fundamental flaws and difficulties associated with the conventional RSS approaches are namely not applicable for time based methods. While several factors severely impacting the RSS values such as transmitter and receiver tilt, LED aging and dust accumulation have a huge impact on the positioning performance, the optical path length and differences remain the same independent of the RSS. Furthermore, it was demonstrated that multipath effects due to reflections have a limited impact on the positioning performance. This implies that the TDoA method is more resilient to changes in the environment compared to typical RSS methods. While the pulse is spread in the time-domain, the other Gaussian pulse fitting parameters A , σ and n are also affected to a noticeable degree, which could enable the detection of the NLoS contributions in future work and facilitate compensation of the impact.

Another significant benefit lies in the calibration only requiring one measurement at a known position, eliminating a meticulous and time-consuming collection of a large training dataset as is typical for RSS models.

VII. CONCLUSION

This work demonstrated the feasibility of TDoA OWP through the utilization of Gaussian pulse fitting. The used hardware and signal processing method was described and evaluated with a high sampling rate oscilloscope. The results demonstrate that an adequate accuracy was achieved for both ranging and

positioning and outperforms cross-correlation. Even with a lower sampling rate, sufficient resolution and accuracy are attainable by employing the Gaussian pulse fitting method. A measurement setup was constructed to evaluate the positioning performance of TDoA on a grid, showing a P95 accuracy of 5.3 cm in a 50×50 cm grid. The calibration of the system was performed only using a single measurement at a known position, constituting a major advantage with regard to practicality compared to acquiring a dataset to train RSS models.

The introduction of a corner geometry around a single receiver produced reflections, resulting in an amplitude increase and time-domain spreading of the received pulses, enlarging the associated ranging errors. However, we have shown that the impact on the positioning accuracy can be largely minimized through a practical calibration.

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